Lab 3 – Architecture Justification

Problem Statement #29 – Municipal Infrastructure Damage Reporting App · SRN PES2UG24AM032

Architecture Selection

We chose a Client-Server architecture (with a layered municipal server) for the Municipal Infrastructure Damage Reporting System. Citizens use a mobile client and department officers use a web dashboard. Both talk to one central municipal server, which holds the Report Manager, Screening Service and Dispatch & Notification components, backed by a single Report & Audit Store.

Alternatives considered

- Layered (monolith only): clear separation, but it has no natural place for an offline-capable mobile client, which NFR-001 requires.
- Microservices: independent scaling and fault isolation, but too much operational overhead (many deployments, distributed data, network latency) for one municipality's report volume and a small IT team. Keeping data consistent across services would also make duplicate detection harder.
- Client-Server: one source of truth with thin, replaceable clients. This matches who uses the system and how much load it gets.

Reason 1 – One source of truth for screening and routing

FR-002 requires automatic checks for duplicate reports, missing photos and out-of-city locations before a report reaches staff. Duplicate detection only works if every new report is compared against all existing reports. A central server with one store makes this a simple query, and FR-003 routing to the correct department uses the same data.

Reason 2 – Thin clients that suit field conditions

Citizens report from roadsides with poor connectivity. The client only captures the geotagged photo and description, queues them offline, and syncs through IReportSubmission when the network returns (NFR-001). All business rules stay on the server, so the mobile app and Officer Dashboard stay small and can be updated without changing the rules.

Security advantage

Every client request passes through one server boundary, so authentication, role-based access (citizen vs. officer vs. engineer on IOfficerWorkflow), input validation and HTTPS are enforced in one place. Status changes are written only by the server to an append-only audit log with the staff ID and timestamp. Clients can't edit or delete history, and a report can't be marked Resolved without photo proof (NFR-002).

Performance benefit

Expensive work is kept off the phone and off the request path. The server screens a report once and hands it to Dispatch & Notification through an asynchronous message queue (IDispatch). The citizen gets an immediate acknowledgement with a ticket number, while routing and push notifications (IStatusNotification) happen in the background. Centralised geocoding results and an indexed store also keep duplicate lookups fast as the number of reports grows.